THE PRIME DIVISORS WHERE A RATIONAL FUNCTION IS NOT INTEGRAL ARE LOCALLY FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The irreducible components of a closed subset are locally finite. Lemma lemma-components-locally-finite, used at proof lines 6650-6651. Concludes the local finiteness.

Source reference. Original lemma 0BE1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14334–14334. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian integral scheme. Let $f \in R(X)^*$. Then the collections*

$$\{Z \subset X \mid Z \text{ a prime divisor with generic point } \xi \text{ and } f \text{ not in } \mathcal{O}_{X,\xi}\}$$

and

$$\{Z \subset X \mid Z \text{ a prime divisor and } ord_Z(f) \neq 0\}$$

are locally finite in X.

Proof. There exists a nonempty open subscheme $U \subset X$ such that f corresponds to a section of $\Gamma(U, \mathcal{O}_X^*)$. Hence the prime divisors which can occur in the sets of the lemma are all irreducible components of $X \setminus U$. Hence Lemma 0BE1 gives the desired result. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.